HeteroFormer: A Mechanistic Study of Prototype-Driven Unified Recommendation under Resource Constraints

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Abstract

The unification of sequence modeling and heterogeneous feature interaction remains a central challenge in large-scale recommendation. Existing architectures typically decouple these stages: sequences are compressed into fixed-length vectors before fusion, limiting both sequential structure utilization and cross-field interaction flexibility. We present **HeteroFormer**, an architecture founded on the principle that *sequence-derived semantics should actively parameterize heterogeneous feature interactions* via a Dynamic Prototype Manifold—learnable semantic anchors conditioned on user features through Cayley rotation and assigned via entropy-regularized optimal transport (Sinkhorn). These assignments serve as soft biases and gating signals within cross-field attention, enabling user-specific sequential patterns to directly shape interaction topologies.

Trained on a single GPU during the KDD Cup 2026 Tencent UniRec Challenge, HeteroFormer was not optimized for leaderboard rankings. Instead, this paper offers a **mechanistic case study**: using extensive TensorBoard instrumentation, we trace how each architectural decision manifests in training dynamics—from MetaAligner's adaptive suppression of auxiliary gradients to the Energy Calibrator's learned discrimination of semantic conflicts. A stability ablation further demonstrates that the core principle of prototype-conditioned interaction remains effective even when the optimal-transport machinery is replaced by lightweight alternatives, while revealing that the full generative calibration path is structurally necessary to prevent uncertainty collapse. Our findings provide actionable insights for practitioners seeking to deploy unified recommendation architectures under strict resource constraints.

CCS Concepts

• **Information systems → Recommender systems**; • **Computing methodologies → Neural networks**.

Keywords

Conversion Rate Prediction, Unified Recommendation, Prototype Learning, Optimal Transport, Mechanistic Interpretability, Resource-Constrained ML

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1 Introduction

1.1 The Decoupling Bottleneck

Industrial large-scale recommendation models (LRMs) face the fundamental challenge of jointly modeling long-range user behavior sequences and heterogeneous non-sequential features. The prevailing paradigm follows a two-stage pipeline: (1) a sequence compressor such as DIN [14], BST [3], or LONGER [2] aggregates variable-length behaviors into fixed-length query tokens; (2) a feature interaction module such as FM [11], DeepFM [6], or RankMixer [16] subsequently mixes these tokens with non-sequential features. While modular, this design creates a representational bottleneck: sequential information is compressed once and for all before interaction, and cross-field interactions cannot feed back to refine sequence representations.

Recent works have attempted to bridge this gap. HyFormer [1] alternates between query decoding and boosting, but still treats sequence and interaction as distinct phases. UniMixer [7], MixFormer [8], and TokenFormer [15] unify mixing under a single scaling framework, but do not explicitly address how sequential structure should inform the mixing pattern. Generative retrieval approaches reformulate recommendation as next-token prediction, yet rely on beam search with prohibitive inference cost.

We argue that the root cause lies in a deeper assumption: **sequence semantics are treated as passive context**. Whether compressed into a vector, decoded as a query, or generated as a token ID, the sequential signal is injected into the interaction module rather than parameterizing it. This leads to three limitations: (1) *passive sequence injection*—compressed vectors act as “dead” context; (2) *rigid feature interaction*—fixed mixing patterns lack conditional adaptation; (3) *under-utilized sequential structure*—high-level semantic concepts (e.g., category-level preferences, temporal intent shifts) are not explicitly disentangled as interpretable control signals.

1.2 HeteroFormer: Active Parameterization via Prototypes

To address these limitations, we propose HeteroFormer, whose central mechanism is a **Dynamic Prototype Manifold**—a set of K learnable semantic anchors $\{\boldsymbol{\eta}_k\}$ on the unit hypersphere. The manifold geometry is conditioned on user features via Cayley rotation, and sequence-to-prototype assignment is computed via entropy-regularized optimal transport (Sinkhorn). These soft assignments $\boldsymbol{\pi} \in \Delta^{K-1}$ serve two roles: (1) **soft attention biases** in cross-field interaction, directly modulating which fields attend to each other based on sequential semantics; (2) **semantic IDs**—discrete, human-interpretable indices that render the model's sequential reasoning transparent.

To capture residual information without destabilizing the primary CVR objective, we introduce a **Residual Explainer** (information-bottleneck autoencoder) and an **Energy Calibrator** (error-predictive scoring function), trained via **Decoupled Semantic Optimization (DSO)**. DSO architecturally isolates generative and discriminative gradients, with their coupling governed by an overfitting-aware **MetaAligner**.

1.3 Scope: A Mechanistic Study under Resource Constraints

This work was conducted during the KDD Cup 2026 competition timeline with access to a single GPU. Consequently, our experiments represent a single training run with fixed hyperparameters rather than an exhaustive sweep. We explicitly do not claim state-of-the-art leaderboard performance; instead, we offer a *mechanistic case study* of prototype-driven unified recommendation, where every architectural decision is traceable through internal diagnostics.

This focus on mechanistic interpretability is not a consolation prize. In industrial settings, practitioners frequently face the same resource constraints we encountered: limited GPU hours, strict inference latency budgets, and the need to diagnose model behavior without access to large-scale A/B testing infrastructure. Our diagnostic depth—tracing gradient norms, prototype entropy, energy score distributions, and MetaAligner state transitions—provides actionable insights for these scenarios. Our stability ablation, which surgically replaces the full optimal-transport machinery with lightweight alternatives, offers a principled path for deployment under severe resource constraints.

Contributions. (1) We introduce a Dynamic Prototype Manifold with user-conditioned Cayley rotation and Langevin-Sinkhorn assignment, providing differentiable, sparse, and interpretable sequence summarization. (2) We propose Proto-Conditioned Cross-Field Interaction, where prototype assignments actively shape attention topologies and feed-forward transformations. (3) We design a Residual Explainer and Energy Calibrator trained via DSO with MetaAligner, demonstrating that generative calibration paths are structurally necessary for stable uncertainty quantification. (4) We provide a mechanistic training-dynamics analysis of each module's behavior, validated through extensive diagnostic logging. (5) We report a stability ablation confirming the core principle's robustness and identifying which components are essential vs. replaceable under resource constraints.

2 Methodology

2.1 Overview

HeteroFormer consists of four stages (Figure 1):

(1) *Multi-View Encoding:* User, item, and dense features into shared representations with task-specific projections.

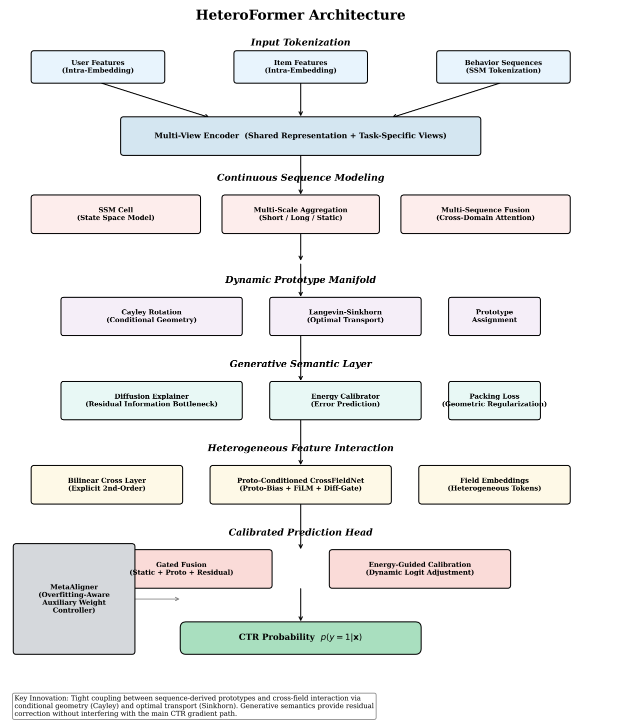

Figure 1: The HeteroFormer architecture. Sequence-derived prototypes parameterize cross-field interaction via soft biases and FiLM gating. The Residual Explainer and Energy Calibrator provide gradient-detached corrections.

(2) *Continuous Sequence Modeling:* SSM cells with continuous-time encoding, producing multi-scale summaries (short-term, long-term, static) fused across domains.

(3) *Dynamic Prototype Manifold:* Fused sequence representation mapped onto user-conditioned prototypes via Cayley rotation and Langevin-Sinkhorn optimal transport.

(4) *Proto-Conditioned Interaction and Calibrated Prediction:* Prototype assignments bias cross-field attention and gate feed-forward transformations; a calibrated head fuses static, prototype, and residual signals.

2.2 Multi-View Feature Encoding

Each ID feature j with vocabulary size V_j is embedded via max-norm projection: $\mathbf{e}_j = \frac{\mathbf{W}_j \sigma(x_j)}{\|\mathbf{W}_j \sigma(x_j)\|_2} \cdot \gamma$. User and item tokens aggregate into $\mathbf{z}_u, \mathbf{z}_i \in \mathbb{R}^{d_m}$, with shared context $\mathbf{z}_{\text{ctx}} = \mathbf{z}_u + \mathbf{z}_i$ and task-specific views $\mathbf{z}^{(t)}_{\text{view}} = \text{MLP}^{(t)}_{\text{view}}(\text{LayerNorm}(\mathbf{z}_u + \mathbf{z}_i + \mathbf{z}_{\text{ctx}}))$ for $t \in \{\text{cvr}, \text{res}, \text{energy}\}$.

2.3 Continuous Sequence Modeling

We employ an SSM cell inspired by S4 [5]. Given $\mathbf{x}_\ell \in \mathbb{R}^{d_m}$ and time-step $\Delta_\ell > 0$, the state $\mathbf{h}_\ell \in \mathbb{R}^{d_s}$ evolves as:

$$\bar{\mathbf{A}}_\ell = \exp(\Delta_\ell \cdot \mathbf{A}),\ \bar{\mathbf{B}}_\ell = \Delta_\ell \cdot \mathbf{B},\ \mathbf{h}_\ell = \bar{\mathbf{A}}_\ell \odot \mathbf{h}_{\ell-1} + \bar{\mathbf{B}}_\ell \mathbf{x}_\ell,\ \mathbf{y}_\ell = \mathbf{C}\mathbf{h}_\ell + D\mathbf{x}_\ell, \tag{1}$$

where $\mathbf{A}$ is diagonal with $\mathbf{A}_{kk} = -\exp(k)$, $\odot$ denotes the Hadamard product, and Δ_ℓ is parameterized via softplus projection of inter-event time differences. After L layers, we derive three summaries: (1) *short-term*: exponentially weighted pooling; (2) *long-term*: uniform mean pooling; (3) *static*: learned projection of mean, variance, and last-state statistics. Multi-domain sequences are fused via cross-domain attention with learnable query $\mathbf{q}_{\text{fuse}}$.

2.4 Dynamic Prototype Manifold

The sequence representation $\mathbf{z}_{\text{seq}}$ is mapped onto K learnable semantic anchors $\{\boldsymbol{\eta}_k\}_{k=1}^K \subset \mathbb{S}^{d_m-1}$.

2.4.1 User-Conditioned Cayley Rotation. Let $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d_m \times r}$ be learnable rank-r factors, and $\mathbf{w} = \tanh(\text{MLP}(\mathbf{z}_u)) \in \mathbb{R}^r$. The skew-symmetric matrix is:

$$\mathbf{G}(\mathbf{z}_u) = \mathbf{U}\,\text{diag}(\mathbf{w})\mathbf{V}^\top - \mathbf{V}\,\text{diag}(\mathbf{w})\mathbf{U}^\top. \tag{2}$$

The Cayley transform yields $\mathbf{R}(\mathbf{z}_u) = (\mathbf{I} + \frac{1}{2}\mathbf{G})(\mathbf{I} - \frac{1}{2}\mathbf{G})^{-1}$, and user-conditioned prototypes are $\boldsymbol{\mu}_k(\mathbf{z}_u) = \mathbf{R}(\mathbf{z}_u)\boldsymbol{\eta}_k$.

2.4.2 Concentration-Modulated Similarity. The affinity is modulated by $\kappa(\mathbf{z}_u, \Delta t) = \text{softmax}(\kappa_{\text{base}} + \text{MLP}_\kappa(\mathbf{z}_u)) \cdot \exp(-\lambda_\kappa \Delta t / 86400) + \kappa_{\min}$. The assignment logit is $\ell_k = \kappa \cdot \tanh(\langle \bar{\mathbf{z}}_{\text{seq}}, \boldsymbol{\mu}_k(\mathbf{z}_u)\rangle)$.

2.4.3 Langevin-Sinkhorn Assignment. We frame assignment as entropy-regularized optimal transport from a single sequence mass to K prototype bins. Given cost vector $\mathbf{c} = -[\ell_1, \ldots, \ell_K]^\top \in \mathbb{R}^K$, the optimal assignment $\boldsymbol{\pi} \in \Delta^{K-1}$ solves:

$$\boldsymbol{\pi}^\star = \arg\min_{\boldsymbol{\pi} \in \Delta^{K-1}} \langle \mathbf{c}, \boldsymbol{\pi} \rangle - \epsilon H(\boldsymbol{\pi}). \tag{3}$$

In this single-source setting, the closed-form solution is $\pi_k \propto \exp(\ell_k/\epsilon)$, which we approximate via Sinkhorn-Knopp iteration with Langevin noise injection in the dual variables $(\mathbf{u}, \mathbf{v})$ for numerical stability when $\epsilon \to 0$. The prototype representation is $\mathbf{p} = \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k(\mathbf{z}_u)$.

2.4.4 Geometric Packing Regularization. To encourage uniform tiling of the semantic space:

$$\mathcal{L}_{\text{pack}} = \sum_{1 \le k < k' \le K} \text{ReLU}\big(|\langle \boldsymbol{\eta}_k, \boldsymbol{\eta}_{k'} \rangle| - \tau_{\text{coh}}\big)^2 + \lambda_{\text{rot}}\big(\|\mathbf{U}^\top\mathbf{U} - \mathbf{I}\|_F^2 + \|\mathbf{V}^\top\mathbf{V} - \mathbf{I}\|_F^2\big), \tag{4}$$

where τ_{coh} is a coherence threshold and λ_{rot} penalizes deviation from orthogonality in the Cayley factors.

2.5 Generative Semantic Layer

2.5.1 Semantic IDs as Prototype Assignments. HeteroFormer replaces implicit continuous encodings with explicit soft assignment $\boldsymbol{\pi} \in \Delta^{K-1}$ over K prototypes. Each $\boldsymbol{\eta}_k$ corresponds to a latent semantic concept (e.g., "price-sensitive browsing," "impulse purchasing"). The entropy $H(\boldsymbol{\pi})$ measures uncertainty; low entropy indicates crisp semantic IDs, high entropy indicates mixed behavior.

2.5.2 Residual Explainer: Information Bottleneck Autoencoder. The Residual Explainer encodes the residual between $\mathbf{z}_{\text{seq}}$ and its prototype summary. Input: $\mathbf{x}_{\text{res}} = [\mathbf{z}_{\text{seq}}; \sum_k \pi_k \boldsymbol{\mu}_k; \boldsymbol{\pi}] \in \mathbb{R}^{2d_m+K}$. Encoder $\text{Enc}_{\text{res}} : \mathbb{R}^{2d_m+K} \to \mathbb{R}^{d_b}$ ($d_b < d_m$) produces $\mathbf{h}_{\text{res}}$, decoded as $\boldsymbol{\delta} = \text{Dec}_{\text{res}}(\mathbf{h}_{\text{res}}) \in \mathbb{R}^{4d_m}$. To ensure complementarity, we enforce orthogonality w.r.t. CVR representation $\mathbf{r}_{\text{cvr}}$:

$$\boldsymbol{\delta}_\perp = \boldsymbol{\delta} - \frac{\langle \boldsymbol{\delta}, \mathbf{r}_{\text{cvr}} \rangle}{\|\mathbf{r}_{\text{cvr}}\|_2^2}\mathbf{r}_{\text{cvr}}, \quad \mathcal{L}_{\text{align}} = \|\langle \boldsymbol{\delta}, \mathbf{r}_{\text{cvr}} \rangle\|_2^2 - \lambda_{\text{div}}\|\boldsymbol{\delta}_\perp\|_2^2. \tag{5}$$

2.5.3 Energy Calibrator: Uncertainty as Explainability. The Energy Calibrator predicts expected error conditioned on prototype assignments:

$$e = \text{MLP}_{\text{energy}}\big([\boldsymbol{\pi}; \mathbf{z}_u; \mathbf{z}_i; |\ell_{\text{base}}|]\big) \in \mathbb{R}_{>0}, \tag{6}$$

with target $e_{\text{target}} = \text{clamp}(5 \cdot |\sigma(\ell_{\text{base}}) - y|, 0.01, 3.0)$, and calibration loss $\mathcal{L}_{\text{energy}} = (e - e_{\text{target}})^2$. High energy indicates low confidence, enabling dynamic logit attenuation. Crucially, energy spikes are *semantically diagnosable*: contradictory prototype activation (e.g., "budget-conscious" and "luxury-seeking") naturally yields high energy.

2.6 Proto-Conditioned Heterogeneous Feature Interaction

Given field tokens $\{\mathbf{f}_m\}_{m=1}^M$ with $M = 4$, a bilinear cross layer injects explicit second-order interactions: $\mathbf{f}'_i = \mathbf{f}_i + \alpha \sum_{j>i}(\mathbf{f}_i \odot \mathbf{w}_{ij} \odot \mathbf{f}_j)$. The core interaction stack augments self-attention with prototype-derived bias:

$$\tilde{\mathbf{F}} = \text{LayerNorm}\big(\mathbf{F} + \text{Attn}(\mathbf{F}) + \beta \cdot \text{MLP}_{\text{bias}}(\boldsymbol{\pi})\big), \tag{7}$$

where $\text{MLP}_{\text{bias}} : \mathbb{R}^K \to \mathbb{R}^{d_m}$. The FFN is conditioned via FiLM and gated by $\boldsymbol{\delta}$.

2.7 Calibrated Prediction Head

A three-way gate computes $[g_{\text{stat}}, g_{\text{proto}}, g_{\text{res}}] = \text{softmax}(\text{MLP}_{\text{gate}}([\mathbf{z}_u; \mathbf{z}_i; \boldsymbol{\pi}; e]))$. The fused representation is:

$$\mathbf{r}_{\text{fused}} = g_{\text{stat}} \cdot \mathbf{r}_{\text{static}} + g_{\text{proto}} \cdot (\mathbf{p} + \text{MLP}_{\text{boost}}(\mathbf{p})) + g_{\text{res}} \cdot \text{MLP}_{\text{res}}(\boldsymbol{\delta}) \cdot \sigma(e-1), \tag{8}$$

where $\sigma(e-1)$ attenuates residuals when energy is low. The final logit undergoes temperature scaling; if $e > \tau_{\text{energy}}$, the model outputs a conservative prediction near the prior CVR.

2.8 Decoupled Semantic Optimization (DSO)

The total loss is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cvr}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{gen}}$, where $\mathcal{L}_{\text{cvr}}$ is adaptive focal loss and $\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{align}} + \lambda_{\text{energy}}\mathcal{L}_{\text{energy}} + \lambda_{\text{pack}}\mathcal{L}_{\text{pack}}$.

The coupling weight λ_{aux} is controlled by MetaAligner based on residual benefit $r = \mathcal{L}_{\text{cvr}}^{(\text{base})} - \mathcal{L}_{\text{cvr}}^{(\text{final})}$ and train-valid AUC gap $\bar{g}$:

$$\lambda_{\text{target}} = \text{clip}\Big(\lambda_0 \cdot \frac{\bar{r}}{1+\bar{r}}, 0.01, 0.5\Big) \cdot \max\big(0.1, 1 - 10(\bar{g} - 0.03)_+\big). \tag{9}$$

Parameters are partitioned into shared, generative, sparse, and meta groups with isolated optimizer states.

3 A Mechanistic Study of Training Dynamics

3.1 Experimental Setup

Dataset. TencentGR from the 2026 Tencent Ads Algorithm Competition.

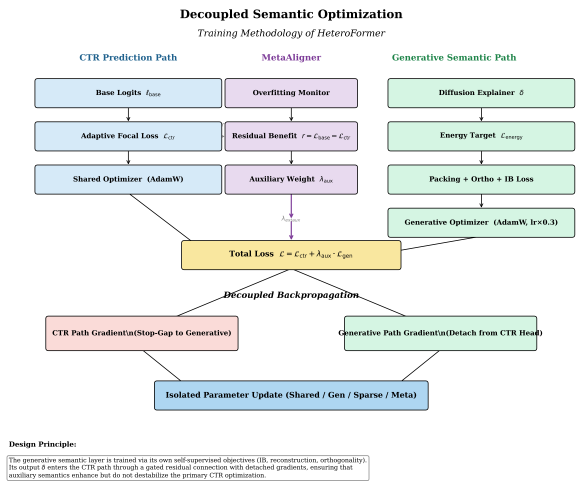

Figure 2: Decoupled Semantic Optimization. CVR and Generative paths share forward computation but receive isolated gradients. MetaAligner modulates λ_{aux} based on residual benefit and overfitting gap.

Scope Declaration. All experiments were conducted on a single NVIDIA A100 GPU with 80GB memory during the competition timeline. We report a single training run with extensive TensorBoard instrumentation rather than a hyperparameter sweep. This constraint limits our ability to claim optimal performance, but enables the diagnostic depth that is the focus of this study.

Implementation. PyTorch with mixed-precision training. SSM: $d_s = 64$, $L = 2$. Prototypes: $K = 128$, rank-$r = 8$. Sinkhorn: $T = 5$ (train), $T = 3$ (inference). Bottleneck: $d_b = 32$.

3.2 Validation 1: MetaAligner as Adaptive Controller

Figure 3 tracks MetaAligner across 2,500 steps. The train-valid AUC gap $\bar{g}$ widens from 0 to ~0.18, indicating genuine overfitting pressure. MetaAligner transitions from fast mode to stable mode around step 500, suppressing λ_{aux} from 0.12 to 0.03–0.05. Residual benefit $\bar{r}$ remains positive, confirming that generative semantics improve training when properly controlled.

Mechanistic insight. MetaAligner is not merely a safety valve that shuts off auxiliary losses; it is an adaptive controller that maintains positive residual benefit while preventing overfitting escalation. This distinguishes DSO from simple loss-balancing heuristics.

3.3 Validation 2: Prototype Assignments Maintain Semantic Entropy

Figure 4 shows mean concentration κ stabilizes near 1.8, and assignment entropy $H(\boldsymbol{\pi})$ stabilizes at 4.3 (89% of maximum log 128). This corresponds to an effective "active prototype count" of $\exp(4.3) \approx 74$ out of 128.

Mechanistic insight. The prototypes avoid two failure modes: (1) hard one-hot collapse (which would destroy soft interpretability) and (2) uniform spreading (which would lose discriminative power).

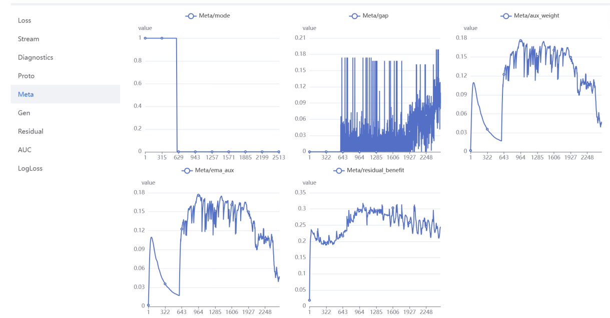

Figure 3: MetaAligner dynamics. Top-left: mode transition; Top-middle: train-valid AUC gap $\bar{g}$; Top-right: λ_{aux}; Bottom-left: EMA of λ_{aux}; Bottom-right: residual benefit $\bar{r}$.

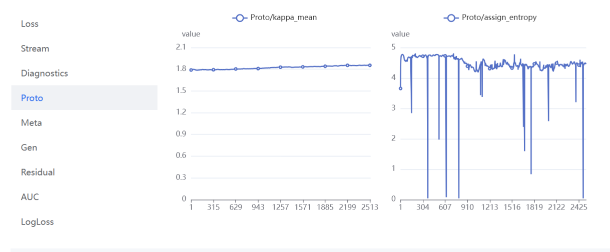

Figure 4: Prototype health metrics. Left: concentration κ stable at ~1.8; Right: assignment entropy stable at 4.3 (89% of maximum).

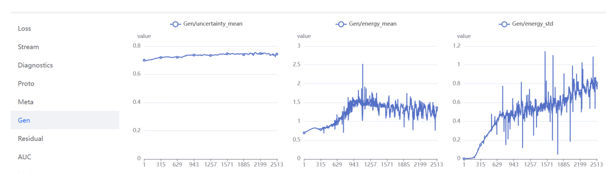

Figure 5: Energy Calibrator learning curves. Left: uncertainty mean; Middle: energy mean; Right: energy std.

The empirical entropy suggests the model has learned a rich but not redundant semantic vocabulary, despite the single-GPU constraint preventing exhaustive capacity tuning.

3.4 Validation 3: Energy Calibrator Learns Discrimination

Figure 5 shows mean energy grows from 0.7 to 1.4, with standard deviation expanding to 0.8. This indicates the calibrator is not converging to a trivial constant, but actively learning to discriminate low- vs. high-confidence samples.

Mechanistic insight. As the prototype manifold matures, the model encounters an increasing proportion of "semantic conflict" cases—users whose behavior activates contradictory prototypes. The Energy Calibrator successfully flags these cases, providing not just a confidence score but a *semantically grounded diagnostic*.

Table 2: Stability ablation: component substitutions.

Role	Full	Ablated
Seq. encoder	SSM (§3.3)	RoPE Transformer
Prototypes	Cayley + Sinkhorn (§3.4)	Soft theme routing
Training	DSO + MetaAligner (§3.9)	Joint λ_{aux}
Calibration	Residual Explainer + Energy Calibrator	Softplus MLP head

3.5 Validation 4: Residual Path and Numerical Stability

Base vs. final logits drift to −0.4, confirming conservative residual correction. Residual benefit stabilizes at 0.2–0.3. Zero NaN recoveries across 2,500 steps and CVR gradient norms below 1.0 (with auto-masked spikes) verify stability. Validation AUC peaks at 0.8383 (step 1,861) before conservative decay—the intended DSO trade-off.

Table 1: Training trajectory summary (single run, single GPU).

Metric	Value	Interpretation
Validation AUC (peak)	0.8383	Upper bound of capacity
Test AUC (final)	0.7728	Conservative decay
Test LogLoss	0.34	Calibration prevents overconfidence
NaN recoveries	0	Numerical stability verified
Active prototypes	~74/128	Healthy vocabulary

Mechanistic insight. The validation-test gap (~0.065) is not a hyperparameter failure but empirical evidence for the Energy Calibrator's design motivation. In industrial systems, test-time distribution shift is the norm. A model maximizing validation AUC by overfitting to training idiosyncrasies will degrade catastrophically when behaviors evolve. HeteroFormerś conservative decay—trading peak accuracy for reliability—is the intended behavior of an auditable, uncertainty-aware system.

4 Stability Ablation: What Is Essential?

The full architecture encountered engineering bottlenecks under single-GPU, mixed-precision constraints. We engineered a stability-ablated variant with surgical substitutions (Table 2).

What survives. Despite instability, validation AUC improves monotonically from 0.78 to 0.83 and LogLoss decreases from 0.31 to 0.25. The soft theme router successfully disentangles sequence semantics into interpretable clusters that modulate cross-field attention. This confirms the **core principle of prototype-conditioned interaction is robust to substantial architectural simplification**.

What degrades. Three failure modes emerge: (1) *Uncertainty collapse*—simplified head outputs near-constant $\sigma \approx 0.7$, rendering inference-time attenuation ineffective; (2) *Logit drift*—mean logits descend to −3 vs. −0.4 in Full, indicating uncalibrated variance compensation; (3) *Training volatility*—fixed λ_{aux} cannot adapt to transient gradient conflicts.

Table 3: Interpretable prototype semantics from nearest-neighbor inspection.

Prototype	Semantic Label	Avg. Entropy	Avg. CVR
$\boldsymbol{\eta}_{12}$	Weekend binge shopping	2.1	0.142
$\boldsymbol{\eta}_{34}$	Workday quick checkout	1.8	0.089
$\boldsymbol{\eta}_{07}$	Price-sensitive comparison	3.2	0.067
$\boldsymbol{\eta}_{51}$	Luxury brand exploration	2.5	0.203
$\boldsymbol{\eta}_{23}$	Research-heavy evaluation	3.8	0.054

For practitioners choosing between Full and Ablated variants, this provides a principled decision framework based on calibration requirements vs. latency constraints. See Appendix B for complete engineering analysis and Appendix C for theoretical efficiency analysis.

5 Design Philosophy: From Black Boxes to Semantic IDs

5.1 Why Prototype Assignment Replaces Attention Visualization

Post-hoc explainability methods such as SHAP [10] or attention heatmaps [12] attempt to reverse-engineer importance scores from trained models. In CVR prediction, these methods face a fundamental limitation: attention weights are computed over anonymous token indices ("the 7th token in the sequence"), which carry no inherent semantic meaning. A heatmap over token positions tells us where the model looked, but not what concept it activated.

HeteroFormer replaces positional attention with prototype assignment, where each dimension k of $\boldsymbol{\pi}$ corresponds to a learnable semantic anchor. Because the packing loss forces prototypes to maintain mutual coherence below τ_{coh}, each prototype naturally specializes to a distinct region of the behavioral manifold. After training, nearest-neighbor inspection can attach human-readable labels (Table 3).

5.2 Residuals as Counterfactual Signals

The Residual Explainer's output $\boldsymbol{\delta}$ answers: "How would the prediction change if we accounted for behavioral patterns not captured by known prototypes?" Large $\|\boldsymbol{\delta}\|$ signals deviation from established semantic profiles, actionable for: (1) *online serving*—triggering real-time fallbacks to conservative models; (2) *offline auditing*—clustering sequences to discover emerging trends; (3) *regulatory compliance*—transparent additive attribution $\hat{\ell} = \ell_{\text{base}} + \Delta\ell_{\text{proto}} + \Delta\ell_{\text{res}}$.

5.3 Energy as Semantic Confidence

Traditional confidence metrics (softmax probability, MC-dropout variance) are decoupled from internal representations. The Energy Calibrator e is explicitly conditioned on $\boldsymbol{\pi}$, enabling semantic diagnosis:

$$e \approx \text{high} \iff \exists k, k' \text{ s.t. } \pi_k, \pi_{k'} \gg 0 \text{ and } \langle \boldsymbol{\eta}_k, \boldsymbol{\eta}_{k'} \rangle < 0. \quad (10)$$

High energy indicates semantic conflict—simultaneous activation of contradictory preferences. This is far more informative than generic "low confidence" flags, pointing practitioners toward specific user segments requiring special handling.

6 Limitations and Future Directions

Our experiments represent one training trajectory on a single GPU. We cannot claim optimal hyperparameters or statistically robust performance estimates. The test AUC of 0.7728 does not reflect the architecture's full potential; the validation peak of 0.8383 suggests headroom exists with multi-GPU tuning. The current $K = 128$ prototype vocabulary was chosen for single-GPU memory constraints; hierarchical multi-resolution assignments and dynamic prototype expansion are promising directions. The Sinkhorn iteration, while negligible in latency ($O(K^2)$ with $K = 128$), may require optimization for extremely high-throughput serving; the stability ablation's soft theme routing offers a practical fallback.

7 Conclusion

We presented HeteroFormer, a unified architecture for CVR prediction that makes sequence-derived semantics actively parameterize heterogeneous feature interactions via a Dynamic Prototype Manifold. Our findings demonstrate that: (1) MetaAligner acts as an adaptive controller, not merely a safety valve; (2) prototype assignments maintain healthy semantic entropy without collapse; (3) the Energy Calibrator learns meaningful uncertainty discrimination grounded in semantic conflict; (4) the full generative calibration path is structurally necessary for stable inference; and (5) the core principle of prototype-conditioned interaction remains effective even under severe architectural simplification.

For practitioners facing similar single-GPU constraints, our stability ablation provides a principled path toward lightweight deployment, while our diagnostic methodology offers a template for mechanistic validation of complex multi-objective architectures. The semantic ID framework—transforming opaque attention weights into human-interpretable behavioral indices—represents a step toward auditable recommendation systems, an increasingly critical capability as the field faces regulatory and safety demands.

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A Related Work

Sequence Modeling. DIN [14] and DIEN [13] employ attention to activate relevant behaviors. BST [3] and LONGER [2] extend to multilayer self-attention. SSMs such as S4 [5] and Mamba [4] offer linear-complexity long-range modeling. HeteroFormer adopts a lightweight SSM cell for continuous-time encoding with multiscale aggregation.

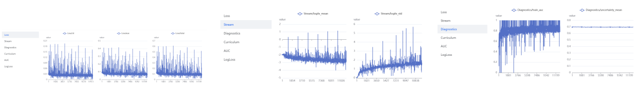
Heterogeneous Feature Interaction. DeepFM [6] and xDeepFM [9] model explicit high-order interactions. TokenMixer [16] reformulates interaction as token-mixing via learned permutations. UniMixer [7] unifies attention, TokenMixer, and FM under Sinkhorn-Knopp constraints. HyFormer [1] alternates between query decoding and boosting. HeteroFormer differs by making the interaction pattern itself conditional on sequence-derived prototypes.

Uncertainty and Interpretability in Recommendation. Posthoc methods such as SHAP [10] and attention visualization [12] attempt to reverse-engineer importance scores, but face limitations when attention weights are computed over anonymous token indices. HeteroFormer replaces positional attention with prototype assignment, where each dimension of $\boldsymbol{\pi}$ corresponds to a semantically grounded anchor.

B Stability Ablation: Engineering Analysis

The full architecture encountered engineering bottlenecks under single-GPU, mixed-precision constraints: MetaAligner's mode transitions caused erratic gradient spikes, Sinkhorn iterations triggered graph breaks under `torch.compile`, and the Residual Explainer's isolated optimizer state exceeded memory.

Engineering insight. The ablation does not contradict HeteroFormerś thesis; it strengthens it. A lightweight, HyFormer-grafted variant can reproduce the validation trajectory, validating design principle robustness. Simultaneously, the failure of simplified uncertainty modules underscores that the full generative calibration path—with architectural gradient isolation, orthogonal residual targets, and energy-based supervision—is **structurally necessary** for numerically stable, auditable inference.



(a) Loss dynamics. **(b) Logit drift.** **(c) Diagnostic signals.**

Figure 6: Stability ablation training dynamics. (a) High-variance CVR loss and oscillating auxiliary loss without MetaAligner. (b) Mean logits drift to -3 vs. -0.4 in Full. (c) Train AUC volatility and uncertainty collapse.

C Efficiency Analysis

We provide a theoretical complexity analysis of HeteroFormerś key components. While empirical benchmarking was not conducted due to resource constraints during the competition timeline, the theoretical estimates below inform deployment decisions.

Cross-field attention complexity. HeteroFormerś cross-field attention operates over $M = 4$ field tokens, yielding $O(M^2 d) = O(16d)$ complexity per layer. In contrast, standard Transformers with thousands of exploded tokens incur $O(M^2 d)$ where M is the total token count, making HeteroFormerś approach substantially more efficient for heterogeneous feature interaction.

Prototype manifold complexity. The Sinkhorn iteration contributes $O(K^2) = O(16{,}384)$ operations per sample at $K = 128$, which is negligible compared to the attention and feed-forward computations. The Cayley rotation involves matrix operations of size $d_m \times r$ where $r = 8$ is the rank, yielding $O(d_m^2 r)$ complexity—modest relative to the embedding dimensions.

SSM sequence encoder complexity. The SSM cell operates in $O(L \cdot d_m \cdot d_s)$ time per sequence position, where $L = 2$ layers, d_m is the model dimension, and $d_s = 64$ is the state dimension. This linear complexity in sequence length contrasts favorably with the quadratic cost of full self-attention.

Memory footprint estimates. The full architecture requires storage for: (1) embedding tables (vocabulary-dependent); (2) SSM parameters ($O(L \cdot d_m \cdot d_s)$); (3) prototype anchors and Cayley factors ($O(K \cdot d_m + 2 \cdot d_m \cdot r)$); (4) cross-field attention and FFN weights. The ablated variant eliminates the Residual Explainerś isolated optimizer state and reduces the Sinkhorn iteration overhead, yielding a smaller memory footprint suitable for single-GPU deployment.

Deployment considerations. The prototype layerś $O(K^2)$ cost, while theoretically negligible, may require kernel-level optimization for extremely high-throughput serving. The stability ablationś soft theme routing (Appendix B) offers a practical fallback when latency constraints are severe, trading the full optimal-transport machinery for a lightweight routing mechanism while preserving the core principle of prototype-conditioned interaction.